

Suicide Mortality in Relation to Dietary Intake of n-3 and n-6 Polyunsaturated Fatty Acids and Fish: Equivocal Findings From 3 Large US Cohort Studies

Intake of n-3 and n-6 polyunsaturated fatty acids (PUFAs) has been implicated in the pathogenesis of depression. We sought to estimate the association between intake of fish and n-3 and n-6 PUFAs and suicide mortality over the course of long-term follow-up. In this prospective cohort study, biennial questionnaires were administered to 42,290 men enrolled in the Health Professionals Follow-up Study (1988–2008), 72,231 women enrolled in the Nurses' Health Study (1986–2008), and 90,836 women enrolled in Nurses' Health Study II (1993–2007). Dietary fish and n-3 and n-6 PUFA intakes were assessed every 4 years using a validated food-frequency questionnaire. Suicide mortality was ascertained through blind physician review of death certificates and hospital or pathology reports. Adjusted relative risks of suicide mortality were estimated with multivariable Cox proportional hazards models and pooled across cohorts using random-effects meta-analysis. The pooled multivariable relative risks for suicide among persons in the highest quartile of intake of n-3 or n-6 PUFAs, relative to the lowest quartile, ranged from 1.08 to 1.46 for n-3 PUFAs ($P_{\text{trend}} = 0.11\text{--}0.52$) and from 0.68 to 1.19 for n-6 PUFAs ($P_{\text{trend}} = 0.09\text{--}0.54$). We did not find evidence that intake of n-3 PUFAs or fish lowered the risk of completed suicide.